

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<tool_call> Reasoning process here :

- Step 1 : Understand the problem .
- The initial cost to plant the tree is \$ 90 .
 - Each year , the tree produces 7 lemons .
 - Each lemon sells for \$ 1 . 5 , so annual revenue = $7 \times 1.5 = \$ 10.50$.
 - Annual cost to water and feed = \$ 3 .
 - So , net profit per year = Revenue - Cost = $10.50 - 3 = \$ 7.50$.

Step 2 : We need to find when Carlos starts earning money , meaning when the cumulative profit (after covering the initial cost) becomes positive .

So , we need to find the number of years , n , such that the total profit after n years exceeds the initial cost of \$ 90 .

Total profit after n years = $n \times \$ 7.50$

We want :
 $7.50n > 90$

Solve for n :
 $n > 90 / 7.50$
 $n > 12$

So , after 12 years , the total profit is $12 \times 7.50 = \$ 90$ â€” exactly covering the initial cost .

Therefore , at the end of year 12 , the cost is fully covered . That means in year 13 , he starts earning a *profit* (since the cost is gone and he 's earning money).

But the question is : " How many years will it take before he starts earning money on the lemon tree ?"

" Starts earning money " means the first year in which he makes a profit *after* covering the initial cost .

So , after 12 years , he breaks even . In year 13 , he starts earning a profit .

Therefore , it will take 13 years before he starts earning money .

But wait â€” is " start earning money " meaning the first year when he has a positive net profit ?

Yes . So even though after 12 years he breaks even , in year 13 he earns money .

So the answer is 13 years .

But let 's double-check : after 12 years , total profit is $12 \times 7.50 = 90$ â€” breaks even .

So , he doesn 't earn money until year 13 .

So the number of years it takes to start earning money is 13 .

< answer > 13 </ answer >